

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

					(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)					12
1	The Modulus and Amplitude of $z = \frac{-1}{2} + i \frac{\sqrt{3}}{2}$ is equal to					1
	a. $r = 1, \theta = \frac{2\pi}{3}$	b. $r = 1, \theta = 2\pi$	c. $r = 1, \theta = 2\pi$	d. None		
2	The real part of $\cos(x + iy)$ is equal to					1
	a. $\sin x \sinh y$	b. $\cos x \cosh y$	c. $-\sin x \sinh y$	d. None		
3	Complex number $z = 1 + i\sqrt{3}$ in polar form is					1
	a. $2 \left\{ \cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right\}$	b. $2 \left\{ \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right\}$	c. $2 \left\{ \cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right\}$	d. None		
4	The order and degree of differential equation $\frac{dy}{dx} = x^2 + y^2$ are					1
	a. (1,1)	b. (3,3)	c. (2,2)	d. None		
5	The integrating factor of the Linear differential equation $\frac{dy}{dx} + (2 \tan x)y = \sin x$					1
	a. $\sec^2 x$	b. $2 \sec^2 x$	c. $3 \sec^2 x$	d. None		
6	The complementary function of $(D^3 - 3D^2 + 4)y = 0$ is					1
	a. $c_1 e^{-x} + (c_2 + c_3 x)e^{2x}$	b. $c_1 e^x + (c_2 + c_3 x)e^{2x}$	c. $c_1 e^x + (c_2 + c_3 x)e^{-2x}$	d. None		
7	The complementary function of $(D^2 - 3D + 2)y = 0$ is					1
	a. $c_1 e^{-x} + c_2 e^{-2x}$	b. $c_1 e^x + c_2 e^{2x}$	c. $c_1 e^{3x} + c_2 e^{2x}$	d. None		
8	The Particular Integral of $(y'' + 4y' + 13)y = 18e^{-2x}$; is					1
	a. e^{-2x}	b. $2 e^{-2x}$	c. $2 e^{2x}$	d. None		

9	The value of a_n in the Fourier series expansion of $f(x) = x^2$ in $(-\pi, \pi)$ is				1
	a. $\frac{4\cos n\pi}{n^2}$	b. $\frac{\cos n\pi}{n^2}$	c. $\frac{2\cos n\pi}{n^2}$	d. None	
10	The value of b_n in the Fourier series expansion of $f(x) = 9 - x^2$ in $(-3, 3)$ is				1
	a. 1	b. 0	c. 2	d. None	
11	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then $\text{curl}\vec{F}$ is				1
	a. $\vec{0}$	b. $\vec{1}$	c. $\vec{2}$	d. None	
12	The vector field along $\vec{F}$ is said to be solenoidal if				1
	a. $\nabla \times \vec{F} = 0$	b. $\nabla \times \vec{F} \neq 0$	c. $\nabla \cdot \vec{F} = 0$	d. None	
Q. 2	Solve the following.				12
A)	If $\tan(A + iB) = x + iy$, prove that (i) $\tan 2A = \frac{2x}{1-x^2-y^2}$ (ii) $\tanh 2B = \frac{2y}{1+x^2+y^2}$.				6
B)	If $\sin(A + iB) = x + iy$, then prove that (i) $\frac{x^2}{\cosh^2 B} + \frac{y^2}{\sinh^2 B} = 1$ (ii) $\frac{x^2}{\sin^2 A} - \frac{y^2}{\cos^2 A} = 1$.				6
Q.3	Solve the following.				12
A)	Solve $x \frac{dy}{dx} + y = x^3 y^6$.				6
B)	Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$.				6
Q. 4	Solve any TWO of the following.				12
A)	Solve $y'' + 4y' + 13y = 18e^{-2x}$; $y(0) = 0$, $y'(0) = 9$.				6
B)	Solve $(D^6 - D^4)y = x^2$.				6
C)	Solve by the method of variation of parameters $(D^2 + 1)y = \sec x \tan x$.				6
Q.5	Solve any TWO of the following.				12
A)	Find the Fourier series of the function $f(x) = x$ in the interval $(0, 2\pi)$.				6
B)	Find Fourier series for $f(x) = 1 - x^2$, $-1 \leq x \leq 1$.				6
C)	Find half range Fourier cosine series for $f(x) = x$, $0 < x < 2$.				6

Q. 6	Solve any TWO of the following.		12
A)	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then show that (i) $\nabla \cdot \vec{r} = 3$ (ii) $\nabla \times \vec{r} = \mathbf{0}$		6
B)	If r and $\vec{r}$ have their usual meanings and $\vec{a}, \vec{b}$ are constant vectors, then prove that (i) $\nabla \cdot (r^n \vec{r}) = (n + 3)r^n$ (ii) $\nabla \times (r^n \vec{r}) = \mathbf{0}$		6
C)	Using Green's theorem, evaluate $\int_C [(y - \sin x)dx + \cos x dy]$ where C is the plane triangle enclosed by the lines $y = 0, x = \frac{\pi}{2}$ and $y = \frac{2}{\pi}x$.		6
	*** End ***		